3 The Maxwell–Bloch equations

The optical Bloch equations describe how a given electric field drives a system of two-level atoms. In this chapter we derive, first of all, the equation which governs the converse action, i.e. how a given atomic polarization generates the radiation field. This equation is obtained by starting from the Maxwell equations and applying a rigorous formulation of the paraxial and slowly varying envelope approximations. The next step is to couple such a wave equation with the optical Bloch equations, thereby obtaining the basic set of equations which describes the radiation–matter interaction. They are called *Maxwell–Bloch equations* in the literature, and a feature of paramount importance is that they constitute a nonlinear system of equations. Finally, we apply them the to the description of two intriguing phenomena, self-induced transparency and superradiance/superfluorescence.

3.1 The Maxwell equations. Paraxial and slowly varying envelope approximations

Let us consider the Maxwell equations in the international system of units MKSA. If we set the free electric charge density ρ and current density $\mathbf{J}$ equal to zero and neglect the magnetization of the medium by setting $\mathbf{B} = \mu_0 \mathbf{H}$ we have

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}, \tag{3.1}$$

$$\nabla \cdot \mathbf{H} = 0, \tag{3.2}$$

$$\nabla \cdot \mathbf{D} = 0, \tag{3.3}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}. \tag{3.4}$$

The $\mathbf{D}$ and $\mathbf{E}$ fields are linked by the constitutive relation

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \tag{3.5}$$

where $\mathbf{P}$ is the polarization density of the medium. By applying the $\nabla \times$ operation to both sides of Eq. (3.1) and taking into account Eqs. (3.4) and (3.5), the identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and the fact that $\epsilon_0 \mu_0 = 1/c^2$ we obtain

$$\nabla^2 \mathbf{E} - \nabla(\nabla \cdot \mathbf{E}) - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathbf{P}}{\partial t^2}. \tag{3.6}$$

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

In general, the atomic polarization has two components [29]; one is the polarization of the two-level atoms that we have considered up to this point and the other is the polarization $\mathbf{P}_{\mathrm{B}}$ of a background material that hosts the two-level atoms. We assume that $\mathbf{P}_{\mathrm{B}}$ depends linearly on the electric field, i.e. $\mathbf{P}_{\mathrm{B}} = \epsilon_0 \chi_{\mathrm{B}} \mathbf{E}$, where χ_{B} is a real constant, i.e. we neglect the linear absorption due to the background medium, which will be considered later in Section 5.4.2. Therefore, in Eqs. (3.5) and (3.6) we must replace $\mathbf{P}$ by $\mathbf{P} + \mathbf{P}_{\mathrm{B}}$, and, defining the background refractive index n_{B} as $n_{\mathrm{B}}^2 = 1 + \chi_{\mathrm{B}}$, we obtain

$$\nabla^2 \mathbf{E} - \nabla(\nabla \cdot \mathbf{E}) - \frac{n_{\mathrm{B}}^2}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathbf{P}}{\partial t^2}, \tag{3.7}$$

while Eq. (3.3) reads

$$\epsilon_0 n_{\mathrm{B}}^2 \nabla \cdot \mathbf{E} = -\nabla \cdot \mathbf{P}. \tag{3.8}$$

Let us assume that the electric field propagates in the positive-z direction (*unidirectional propagation*),

$$\mathbf{E}(x, y, z, t) = \frac{1}{2} \left[\mathbf{E}_0(x, y, z, t) \mathrm{e}^{-i(\omega_0 t - k_0 z)} + \text{c.c.} \right], \tag{3.9}$$

where ω_0 is the central frequency of the field, $k_0 = \omega_0 n_{\mathrm{B}}/c$, and $\mathbf{E}_0$ is the field envelope, which varies in space and time much more slowly than the carrier factor $\mathrm{e}^{-i(\omega_0 t - k_0 z)}$. Similarly, we will write

$$\mathbf{P}(x, y, z, t) = \frac{1}{2} \left[\mathbf{P}_0(x, y, z, t) \mathrm{e}^{-i(\omega_0 t - k_0 z)} + \text{c.c.} \right]. \tag{3.10}$$

We can expand $\mathbf{E}$ on the plane-wave basis

$$\mathbf{e}_{\mathbf{k}}^{(i)} \mathrm{e}^{i \mathbf{k} \cdot \mathbf{x}}, \qquad \mathbf{k} \equiv (k_x, k_y, k_z), \qquad i = 1, 2, 3, \tag{3.11}$$

where $\mathbf{e}_{\mathbf{k}}^{(i)}$ are the three orthogonal unit vectors with $\mathbf{e}_{\mathbf{k}}^{(3)} = \mathbf{k}/k$. The *paraxial approximation* consists in assuming that

$$|k_x|, \ |k_y| \ll k_z \sim k_0. \tag{3.12}$$

In the following we show that, even if the condition $\nabla \cdot \mathbf{E} = 0$ does not hold exactly, it becomes valid in the limit of the paraxial approximation and of the slowly varying envelope approximation that we define below. Hence we discard the unit vector $\mathbf{e}_{\mathbf{k}}^{(3)}$ and keep only $\mathbf{e}_{\mathbf{k}}^{(i)}$, $i = 1, 2$, which are orthogonal to $\mathbf{k}$. In this approximation the component of $\mathbf{E}$ in the z direction is much smaller than the orthogonal components because the condition $\nabla \cdot \mathbf{E} = 0$ applied to the plane wave (3.11) with $i = 1, 2$ gives

$$\left| \left(\mathbf{e}_{\mathbf{k}}^{(i)} \right)_z \right| = \left| \frac{k_x}{k_z} \left(\mathbf{e}_{\mathbf{k}}^{(i)} \right)_x + \frac{k_y}{k_z} \left(\mathbf{e}_{\mathbf{k}}^{(i)} \right)_y \right| \ll 1. \tag{3.13}$$

We can write

$$\mathbf{E}(x, y, z, t) = \frac{1}{2} [\hat{\mathbf{e}} E_{0\mathrm{T}}(x, y, z, t) + \hat{\mathbf{e}}_z E_{0\mathrm{L}}(x, y, z, t)] \mathrm{e}^{-i(\omega_0 t - k_0 z)} + \text{c.c.}, \tag{3.14}$$

$$\mathbf{P}(x, y, z, t) = \frac{1}{2} [\hat{\mathbf{e}} P_{0\mathrm{T}}(x, y, z, t) + \hat{\mathbf{e}}_z P_{0\mathrm{L}}(x, y, z, t)] \mathrm{e}^{-i(\omega_0 t - k_0 z)} + \text{c.c.}, \tag{3.15}$$

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

where $\hat{\mathbf{e}}_z$ is the unit vector of the z axis and $\hat{\mathbf{e}}$ is a unit vector orthogonal to it, T indicates transverse and L longitudinal.

In this section we will follow a simplified version of the approach of Lax [30, 31]. The fields are characterized by three spatial scales: the wavelength in the background material $\lambda = 2\pi/k_0$ and the lengths d_0 and L over which the envelopes vary appreciably in the transverse and longitudinal directions, respectively. If we take into account that k_x, $k_y \sim d_0^{-1}$ the paraxial approximation (3.12) amounts to the condition

$$\frac{\lambda}{d_0} \ll 1. \tag{3.16}$$

On the other hand, the so-called *slowly varying envelope approximation* prescribes that the field amplitude varies in z and t much more slowly than the carrier wave, i.e. that

$$\left|\frac{\partial \mathbf{E}_0}{\partial z}\right| \ll k_0|\mathbf{E}_0|, \qquad \left|\frac{\partial \mathbf{E}_0}{\partial t}\right| \ll \omega_0|\mathbf{E}_0|, \tag{3.17}$$

hence we have

$$\frac{\lambda}{L} \ll 1. \tag{3.18}$$

More quantitative conditions can be obtained if, as is reasonable, we assume that L and d_0 are linked by the same (apart from a factor of 2, see Eq. (26.6)) relation as the diffraction length and the beam width in the paraxial approximation, i.e. that

$$L = k_0 d_0^2 \quad \Rightarrow \quad d_0 = \left(\frac{\lambda L}{2\pi}\right)^{1/2}. \tag{3.19}$$

We can then introduce a smallness parameter ϵ such that

$$\epsilon = \frac{1}{k_0 d_0} = \frac{d_0}{L} \quad \Rightarrow \quad \frac{1}{k_0 L} = \epsilon^2, \tag{3.20}$$

which agrees with, but is more precise than, inequality (3.18).

If we now assume that $P_{0\mathrm{T}}/(\epsilon_0 E_{0\mathrm{T}})$, $P_{0\mathrm{L}}/(\epsilon_0 E_{0\mathrm{L}}) \sim \epsilon^2$, as will be confirmed by our subsequent analysis, then from Eq. (3.8) it follows that the condition $\nabla \cdot \mathbf{E} = 0$ (which holds exactly for a linear medium) is approached in the limit $\epsilon \to 0$. Moreover, in the limit $\epsilon \ll 1$ a simple relation that links $E_{0\mathrm{L}}$ with $E_{0\mathrm{T}}$ can be found because if we divide the ∇ operator into a transverse part and a longitudinal part,

$$\nabla = \nabla_\perp + \hat{\mathbf{e}}_z \frac{\partial}{\partial z}, \qquad \nabla_\perp = \hat{\mathbf{e}}_x \frac{\partial}{\partial x} + \hat{\mathbf{e}}_y \frac{\partial}{\partial y}, \tag{3.21}$$

and apply the slowly varying envelope approximation, so that $\partial E_z/\partial z \propto i k_0 E_{0\mathrm{L}}$, the condition $\nabla \cdot \mathbf{E} = 0$ is equivalent to

$$E_{0\mathrm{L}} = \frac{i}{k_0} \nabla_\perp \cdot (\hat{\mathbf{e}} E_{0\mathrm{T}}), \tag{3.22}$$

which implies that $|E_{0\mathrm{L}}/E_{0\mathrm{T}}| \sim (k_0 d_0)^{-1} = \epsilon \ll 1$.

Even though we have used the condition $\nabla \cdot \mathbf{E} = 0$ to determine Eq. (3.22) for $E_{0\mathrm{L}}$, we insert into Eq. (3.7) the exact relation (3.8) in order to be able to evaluate exactly the order of magnitude of all terms for $\epsilon \ll 1$. Thus, we obtain the equation

$$\nabla^2 \mathbf{E} - \frac{n_B^2}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2 \mathbf{P}}{\partial t^2} - \frac{1}{\epsilon_0 n_B^2}\nabla(\nabla \cdot \mathbf{P}). \tag{3.23}$$

By inserting Eqs. (3.14) and (3.15), equating separately the terms containing $e^{-i(\omega_0 t - k_0 z)}$ and dividing the result by the exponential factor, we obtain the following equation for the transverse parts directed along $\hat{\mathbf{e}}$:

$$\begin{aligned}
&\nabla_\perp^2 E_{0T} + 2ik_0 \frac{\partial E_{0T}}{\partial z} + \frac{\partial^2 E_{0T}}{\partial z^2} + 2i\frac{k_0 n_B}{c}\frac{\partial E_{0T}}{\partial t} - \frac{n_B^2}{c^2}\frac{\partial^2 E_{0T}}{\partial t^2} \\
&= -\frac{1}{\epsilon_0 n_B^2}\left\{k_0^2 P_{0T} + 2i\frac{k_0 n_B}{c}\frac{\partial P_{0T}}{\partial t} - \frac{n_B^2}{c^2}\frac{\partial^2 P_{0T}}{\partial t^2}\right. \\
&\qquad \left. + \nabla_\perp\left[\nabla_\perp \cdot (\hat{\mathbf{e}} P_{0T}) + \left(ik_0 + \frac{\partial}{\partial z}\right) P_{0L}\right]\right\},
\end{aligned} \tag{3.24}$$

where we have taken into account that $\omega_0 = ck_0/n_B$ and the transverse Laplacian $\nabla_\perp^2$ is defined as

$$\nabla_\perp^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \tag{3.25}$$

If we now divide both terms of Eq. (3.24) by $2ik_0$, and introduce the scaled dimensionless variables

$$\bar{x} = \frac{x}{d_0}, \quad \bar{y} = \frac{y}{d_0}, \quad \bar{z} = \frac{z}{L}, \quad \bar{t} = \frac{ct}{n_B L}, \tag{3.26}$$

we can recast Eq. (3.24) as follows:

$$\begin{aligned}
&\frac{1}{2i}\bar{\nabla}_\perp^2 E_{0T} + \frac{\partial E_{0T}}{\partial \bar{z}} + \frac{\partial E_{0T}}{\partial \bar{t}} + \frac{\epsilon^2}{2i}\left(\frac{\partial^2 E_{0T}}{\partial \bar{z}^2} - \frac{\partial^2 E_{0T}}{\partial \bar{t}^2}\right) \\
&= \frac{1}{2\epsilon_0 n_B^2}\left\{\frac{i}{\epsilon^2} P_{0T} - 2\frac{\partial P_{0T}}{\partial \bar{t}} - i\epsilon^2 \frac{\partial^2 P_{0T}}{\partial \bar{t}^2}\right. \\
&\qquad \left. + i\,\bar{\nabla}_\perp\left[\bar{\nabla}_\perp \cdot (\hat{\mathbf{e}} P_{0T}) + \frac{1}{2}\left(1 + \epsilon^2 \frac{\partial}{\partial \bar{z}}\right) P_{0L}\right]\right\}.
\end{aligned} \tag{3.27}$$

From this we see that, consistently with our initial assumption, the ratio $P_{0T}/(\epsilon_0 E_{0T})$ is of order ϵ^2. Similarly, from the equation for the longitudinal part (which we do not write explicitly here) we obtain also that $P_{0L}/(\epsilon_0 E_{0L}) \sim \epsilon^2$, and since $E_{0L}/E_{0T} \sim \epsilon$ it follows that $P_{0L}/P_{0T} \sim \epsilon$ and $P_{0L}/\epsilon_0 E_{0T} \sim \epsilon^3$. Hence, the fields coincide with their transverse parts to leading order and we write

$$\mathbf{E}(x, y, z, t) = \frac{\hat{\mathbf{e}}}{2} E_0(x, y, z, t) e^{-i(\omega_0 t - k_0 z)} + \text{c.c.}, \tag{3.28}$$

$$\mathbf{P}(x, y, z, t) = \frac{\hat{\mathbf{e}}}{2} P_0(x, y, z, t) e^{-i(\omega_0 t - k_0 z)} + \text{c.c.} \tag{3.29}$$

The final equation for E_0 and P_0 is obtained from Eq. (3.27) by dropping the terms proportional to ϵ^2 and ϵ^4 and the suffix T, coming back to the original variables x, y, z and

t, and taking into account that $L = k_0 d_0^2$:

$$\frac{1}{2ik_0}\nabla_\perp^2 E_0 + \frac{\partial E_0}{\partial z} + \frac{n_B}{c}\frac{\partial E_0}{\partial t} = i\frac{k_0}{2\epsilon_0 n_B^2}P_0. \tag{3.30}$$

We note in particular that no term originating from $\nabla(\nabla \cdot \mathbf{E})$ survives in Eq. (3.30).

3.2 The Maxwell–Bloch equations. The plane-wave approximation

At this point we can combine the wave equation (3.30) with the optical Bloch equations (2.46) and (2.47) in a consistent way. In the wave equation we note the slowly varying envelopes of the field E_0 and of the macroscopic polarization density P_0; in the Bloch equations we note the Rabi frequency Ω and the slowly varying envelope of the microscopic polarization r. Since, in accord with Eqs. (2.39) and (2.42) one has $\Omega = dE_0/\hbar$ and $P_0 = 2Ndr$, where N is the atomic density, we obtain the closed set of equations

$$\frac{1}{2ik_0}\nabla_\perp^2 \Omega + \frac{\partial \Omega}{\partial z} + \frac{n_B}{c}\frac{\partial \Omega}{\partial t} = i\frac{k_0}{\epsilon_0 n_B^2}N\frac{d^2}{\hbar}r, \tag{3.31}$$

$$\frac{\partial r}{\partial t} = -i\delta r - \frac{i}{2}\Omega r_3, \tag{3.32}$$

$$\frac{\partial r_3}{\partial t} = i(\Omega r^* - \Omega^* r), \tag{3.33}$$

where Ω, r and r_3 depend on the variables x, y, z and t. These are the *Maxwell–Bloch equations* [32, 33] which govern the interaction of a system of two-level atoms with the electric field. Such equations hold in the (i) dipole, (ii) rotating wave, (iii) paraxial and (iv) slowly varying envelope approximations.

It is of fundamental importance to observe that, when the Rabi frequency is not a given function of time and space but rather is a variable coupled to the atomic variables by the Maxwell–Bloch equations, the dynamics is nonlinear because Eqs. (3.32) and (3.33) contain products of variables.

In the remainder of this chapter we will also use the *plane-wave approximation*, which consists in assuming that the functions Ω, r and r_3 depend only on the longitudinal variable z and on time t and are independent of the transverse variables x and y. Thus, the field equation reduces to

$$\frac{\partial \Omega}{\partial z} + \frac{n_B}{c}\frac{\partial \Omega}{\partial t} = i\frac{k_0}{\epsilon_0 n_B^2}N\frac{d^2}{\hbar}r. \tag{3.34}$$

We note that the combination of the paraxial plus slowly varying envelope approximation and the plane-wave approximation introduces a drastic simplification because it leads to a dynamical model in which all the equations contain only first-order derivatives.

3.3 Self-induced transparency, the sine–Gordon equation and solitons

There are conditions such that a light pulse propagates along a medium without modifying its shape. Therefore the medium turns out to be transparent to the pulse, and one calls this phenomenon, which was predicted and experimentally demonstrated by McCall and Hahn [34], *self-induced transparency*. In order that this may happen it is necessary that the pulse area defined by Eq. (2.79) in the limit $t \to +\infty$ becomes equal to 2π. In this way the passage of the pulse causes a 2π rotation of the Bloch vector and leaves the medium in the same state as it was in before the pulse.

Let us consider[1] the Maxwell–Bloch equations (3.34), (3.32) and (3.33) in the resonant case and let us re-express the Bloch equations in terms of the quantities r_1 and r_2 defined by Eqs. (2.50) and (2.51). Since $\delta = 0$, r_1 is constant in time, and, if we assume that initially the atoms are in the lower state for all z, then $r_1 = 0$ for all values of time and space and therefore we can ignore it. Hence the Maxwell–Bloch equations read (setting $n_{\mathrm{B}} = 1$ for the sake of simplicity)

$$c\frac{\partial \Omega}{\partial z} + \frac{\partial \Omega}{\partial t} = \beta^2 r_2, \tag{3.35}$$

$$\frac{\partial r_2}{\partial t} = \Omega r_3, \tag{3.36}$$

$$\frac{\partial r_3}{\partial t} = -\Omega r_2, \tag{3.37}$$

where

$$\beta^2 = \omega_0 N \frac{d^2}{2\epsilon_0 \hbar}, \tag{3.38}$$

is a parameter with dimensions equal to those of a squared frequency. The Rabi-frequency envelope remains real if it is real initially, and the motion of the Bloch vector occurs over the circle of unit radius in the (r_2, r_3) plane. We now express the variables r_2 and r_3 as a function of the tipping angle θ in Fig. 2.2 as in Eq. (2.77), so that we obtain an equation essentially identical to Eq. (2.78),

$$\frac{\partial \theta}{\partial t} = \Omega(z, t), \tag{3.39}$$

where we take $t = -\infty$ as the initial time, so that $\theta(-\infty) = 0$. If we insert the equation $r_2 = -\sin\theta$ into Eq. (3.35) we obtain

$$c\frac{\partial \Omega}{\partial z} + \frac{\partial \Omega}{\partial t} = -\beta^2 \sin\theta. \tag{3.40}$$

[1] The following treatment in this section is inspired by lectures delivered by Rodolfo Bonifacio.

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

Next, taking Eq. (3.39) into account, we arrive at a closed equation for the tipping angle θ:

$$c\frac{\partial^2\theta}{\partial z\,\partial t}+\frac{\partial^2\theta}{\partial t^2}=-\beta^2\sin\theta. \tag{3.41}$$

If we transform to a retarded time, i.e. we set

$$\tau_1=t-\frac{z}{c},\quad z_1=z, \tag{3.42}$$

Eq. (3.41) takes the form

$$c\frac{\partial^2\theta}{\partial z_1\,\partial\tau_1}=-\beta^2\sin\theta, \tag{3.43}$$

which is called the *sine–Gordon equation*.[2] Let us now look for a solution of Eq. (3.41) with the form of a pulse that propagates along z with arbitrary velocity v, i.e. $\theta(t-z/v)$. If we define $\tau=t-z/v$ we obtain from Eq. (3.41) or from Eq. (3.43) the following equation:

$$\frac{d^2\theta}{d\tau^2}=\beta'^2\sin\theta,\qquad \beta'^2=\frac{\beta^2}{c/v-1}, \tag{3.44}$$

where we have assumed that $v<c$. This is a pendulum equation, if θ is measured from the unstable equilibrium point. Let us now observe that, if θ is a solution of the equation

$$\frac{d\theta}{d\tau}=2\beta'\sin\left(\frac{\theta}{2}\right), \tag{3.45}$$

then it is also a solution of Eq. (3.44), since if we derive Eq. (3.45) we obtain

$$\frac{d^2\theta}{d\tau^2}=\beta'\cos\left(\frac{\theta}{2}\right)\frac{d\theta}{d\tau}=2\beta'^2\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)=\beta'^2\sin\theta. \tag{3.46}$$

The fact that θ is a solution of Eq. (3.45) implies that with the initial condition $\theta(-\infty)=0$ there is associated also the other initial condition $\dot\theta(-\infty)=0$. In order to find the solution of Eq. (3.45) let us introduce the substitution $\theta=4\arctan x$, from which it follows that

$$\frac{d\theta}{d\tau}=\frac{4}{1+x^2}\frac{dx}{d\tau}. \tag{3.47}$$

[2] This name originates from the equation

$$\frac{\partial^2\theta}{\partial t^2}-c^2\frac{\partial^2\theta}{\partial z^2}=-\beta^2\sin\theta,$$

which, if one replaces the $\sin\theta$ term by the linear term θ, reduces to the one-dimensional Klein–Gordon equation. If, in such an equation, one introduces the transformation

$$\tau_1=t-\frac{z}{c},\quad z_1=z+ct,$$

the equation takes the form

$$4c\frac{\partial^2\theta}{\partial z_1\,\partial\tau_1}=-\beta^2\sin\theta,$$

which, apart from a numerical factor, coincides with Eq. (3.43).

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

Furthermore, since $\sin(2\alpha) = 2\tan\alpha/(1+\tan^2\alpha)$ we have also that

$$\sin\left(\frac{\theta}{2}\right) = \sin(2\arctan x) = \frac{2x}{1+x^2}, \tag{3.48}$$

hence the equation for x is simply $dx/d\tau = \beta' x$ from which

$$x = A\mathrm{e}^{\beta'\tau}. \tag{3.49}$$

The circumstance that $x(-\infty) = 0$ is in accord with the fact that $x = \tan(\theta/4)$ and $\theta(-\infty) = 0$. We have also that $x(+\infty) = +\infty$, and this implies that $\theta(+\infty) = 2\pi$, i.e. the pulse area, is equal to 2π.[3] Finally, if we assume that the pulse is symmetrical in τ and therefore $\theta(0) = \pi$ from which $x(0) = 1$, we have that $A = 1$. Therefore, using Eq. (3.48), the solution of Eq. (3.44) has the form

$$\sin\left(\frac{\theta}{2}\right) = \frac{2x}{1+x^2} = \frac{2\mathrm{e}^{\beta' t}}{1+\mathrm{e}^{2\beta' t}} = \frac{1}{\cosh(\beta'\tau)} = \mathrm{sech}(\beta'\tau). \tag{3.50}$$

From Eq. (3.45) it follows that

$$\Omega(z,t) = \frac{\partial\theta}{\partial t} = \frac{d\theta}{d\tau} = 2\beta'\,\mathrm{sech}(\beta'\tau) = 2\beta'\,\mathrm{sech}\left[\beta'\left(t - \frac{z}{v}\right)\right]. \tag{3.51}$$

This equation describes a pulse that propagates with constant velocity while conserving its shape. Such pulses are called *self-induced transparency solitons*. If we call $\tau_0 = 1/\beta'$ the pulse width, we have

$$\Omega(z,t) = \frac{2}{\tau_0}\,\mathrm{sech}\left(\frac{\tau}{\tau_0}\right) = \frac{4}{\tau_0}\frac{1}{\mathrm{e}^{\tau/\tau_0} + \mathrm{e}^{-\tau/\tau_0}}, \tag{3.52}$$

which shows that the pulse is higher the shorter it is, in accord with the fact that the area must be equal to 2π. The propagation velocity is given by the relation $\beta'^2 = \beta^2/(c/v - 1)$, from which

$$v = \frac{c}{1+\beta^2\tau_0^2}, \tag{3.53}$$

in accord with the condition $v < c$. The pulse is faster the shorter it is. One can show that the solitons can collide, because a faster soliton can overtake a slower one, but at the end of the collision they assume their original shapes again. Figure 3.1 shows the soliton for various values of τ_0 and also the corresponding profile for the inversion r_3 which, taking Eqs. (2.77) and (3.50) into account, is given by

$$r_3 = \frac{6 - \mathrm{e}^{-2\beta'\tau} - \mathrm{e}^{2\beta'\tau}}{2 + \mathrm{e}^{-2\beta'\tau} + \mathrm{e}^{2\beta'\tau}}. \tag{3.54}$$

The energy density, and therefore the radiation intensity, is proportional to the square of the electric field and, taking Eq. (3.9) into account and averaging over time, is proportional to $|E_0|^2$ and therefore to $|\Omega|^2$. Hence in Fig. 3.1 we plot Ω^2 (Ω is real on resonance).

[3] In principle the pulse area might be equal to any integer multiple of 2π, but one can demonstrate that pulses of area $2n\pi$ with $n > 1$ are unstable and decay into n pulses of area 2π.

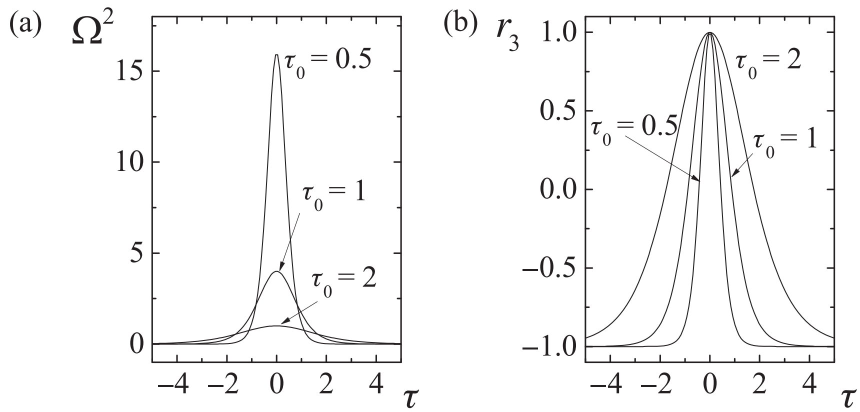

Figure 3.1 (a) The profile of the soliton intensity for some values of τ_0. (b) The profile of the inversion for the same values of τ_0.

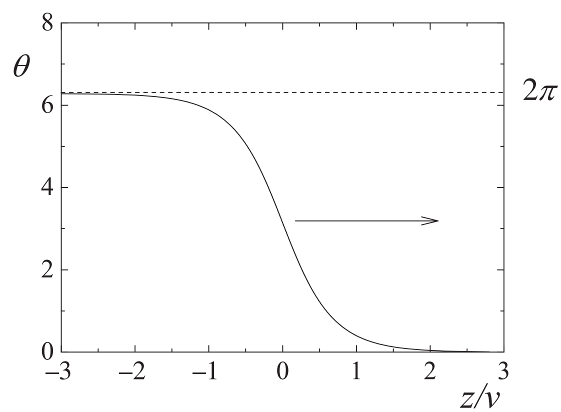

Figure 3.2 A plot of the function $\theta(\tau)$ for $t = 0$, $\tau_0 = 1$. The arrow indicates how the kink moves with time.

Since from Eq. (3.48, 3.49)

$$\theta(\tau) = 4 \arctan x(\tau) = 4 \arctan e^{\beta'(t-z/v)}, \tag{3.55}$$

the graph of θ as a function of z for a fixed value of t has the kink structure shown in Fig. 3.2.

3.4 Superradiance and superfluorescence

In 1954 Dicke [35] (see also [36]) predicted that a system of many two-level atoms, contained in a small region of volume smaller than the wavelength to the third power and prepared in such a way that the atomic system has a macroscopic dipole moment, emits a radiation pulse with intensity proportional to the square of the number of atoms, instead of the standard emission with intensity proportional to the number of atoms, and he called this phenomenon *superradiance*. This remarkable behavior arises from the cooperative

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

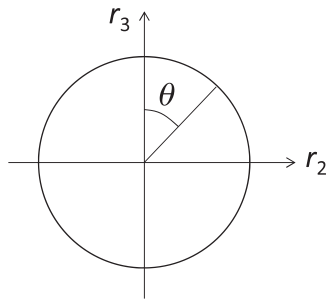

Figure 3.3 The definition of the tipping angle θ used for superradiance/superfluorescence.

character of the emission, which is the coherent emission from a system of aligned dipoles. Subsequent investigations considered a system of atoms that, instead of being point-like, occupies a pencil-shaped region, studying the propagation of the emitted light along the sample and predicting that also in this case one has superradiant emission.

In the 1970s Bonifacio, Schwendimann and Haake [37] considered the case in which the atomic system has initially no macroscopic dipole, but is in the upper state, and showed that also in this case there can be the emission of a pulse with intensity proportional to the square of the number of atoms. The atomic dipoles are initially in a disordered state, but nonetheless they realize a sort of self-organization that produces the *cooperative emission*. While in the case of superradiance the phenomenon is classical because the atomic system emits just like a classical antenna, in the case of initially excited atoms the emission starts as spontaneous emission and therefore needs a quantum description. Since in standard fluorescence (spontaneous emission) the intensity is simply proportional to the number of atoms, Bonifacio introduced later the name *superfluorescence* to designate the cooperative spontaneous emission with peak intensity proportional to the square of the number of atoms. It is good to keep in mind that in the literature the name superfluorescence is used also for kinds of amplified spontaneous emission in which the proportionality to the square of the number of atoms is missing, but here we will stick to the definition of superfluorescence given above.

Again, let us assume exact resonance between the radiation field and the atoms ($\delta = 0$) and let us start from the Maxwell–Bloch equations (3.35)–(3.37) for a pencil-shaped sample of length L. In the case of superradiance/superfluorescence it is convenient to consider a tipping angle θ measured from the North pole of the Bloch circle as in Fig. 3.3, so that

$$r_2(z,t) = \sin\theta(z,t), \qquad r_3(z,t) = \cos\theta(z,t), \tag{3.56}$$

and Eq. (3.39) remains unaltered, whereas Eq. (3.40) becomes

$$c\frac{\partial\Omega}{\partial z} + \frac{\partial\Omega}{\partial t} = +\beta^2 \sin\theta, \tag{3.57}$$

where, again, we have set $n_B = 1$. At this stage, we introduce an approximation that is rather crude but allows one to perform an analytical treatment and to capture the essential features of the phenomena in play, leading to results in qualitative agreement with the experimental findings. The propagation term $c\,\partial\Omega/\partial z$ represents a loss because the radiation escapes

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

from the edge $z = L$ of the pencil-shaped sample. The approximation consists in replacing this term by a damping term $k\Omega$, with $k = c/L$, and in neglecting the spatial variation by considering directly the field emitted at $z = L$, so that Eq. (3.57) becomes

$$\dot{\Omega} + k\Omega = \beta^2 \sin\theta, \tag{3.58}$$

where we have written $\dot{\Omega}$ because now the problem is in time only. As we will see in Chapter 12, this step becomes rigorous when the atomic system is contained in an optical cavity of high quality. Using Eq. (3.39) we obtain the following damped pendulum equation for the tipping angle:

$$\ddot{\theta} + k\dot{\theta} = \frac{1}{\tau_c^2} \sin\theta, \tag{3.59}$$

where we have introduced the parameter $\tau_c = 1/\beta$, which in the literature on superradiance/superfluorescence is called the cooperation time. It was introduced by Arecchi and Courtens [38].

An initial condition for Eq. (3.59) is $\dot{\theta} = 0$ because the electric field is zero at the beginning of the emission. The initial value $\theta(0) = \theta_0$ of the tipping angle is different for superradiance and superfluorescence. In the case of superradiance one has $r_2(0) = 1$, $r_3(0) = 0$ and therefore $\theta_0 = \pi/2$; in the case of superfluorescence one has $r_2(0) = 0$, $r_3(0) = 1$ and therefore $\theta_0 = 0$. However, the state $\theta(0) = 0$, $\dot{\theta}(0) = 0$ corresponds to a pendulum starting from the unstable equilibrium state with zero velocity and therefore the solution of Eq. (3.59) would be the (unstable) stationary solution $\theta(t) = 0$, i.e. the emission would not start. This is due to the fact that superfluorescent emission is initiated by spontaneous emission in the direction of the z axis, which is missing in the semiclassical picture. Superfluorescence needs a quantum picture to describe its initiation. The quantum-mechanical analysis of [37] shows that, in the semiclassical picture, the quantum initiation can be simulated by taking for the tipping angle the initial value $\theta(0) = \sqrt{2/\mathcal{N}}$, where $\mathcal{N}$ is the number of atoms in the pencil-shaped sample. This small initial deviation from the unstable equilibrium point allows the emission to start. Hence the initial conditions are

$$\theta(0) = \theta_0 = \frac{\pi}{2}, \quad \dot{\theta}(0) = 0 \tag{3.60}$$

for superradiance and

$$\theta(0) = \theta_0 = \sqrt{\frac{2}{\mathcal{N}}}, \quad \dot{\theta}(0) = 0 \tag{3.61}$$

for superfluorescence. The most typical configuration for superradiance/superfluorescence is that defined by the condition $k\tau_c \gg 1$ (*pure superradiance/superfluorescence*). In this limit the pendulum described by Eq. (3.59) is overdamped, i.e. approaches the stable equilibrium state $\theta = \pi$ without oscillations, and the inertial term $\ddot{\theta}$ can be dropped, so the pendulum equation reduces to

$$\dot{\tilde{\theta}} = \frac{2}{\tau_R} \sin\left(\frac{\tilde{\theta}}{2}\right), \tag{3.62}$$

https://doi.org/10.1017/CBO9781107477254.005 Published online by Cambridge University Press

where we have set $\tilde{\theta} = 2\theta$ and introduced the superradiance/superfluorescence time τ_R defined as

$$\tau_R = k\tau_c^2 \longrightarrow k\tau_R = (k\tau_c)^2. \tag{3.63}$$

Note that the condition $k\tau_c \gg 1$ implies that $k\tau_R \gg 1$. In addition, if we introduce the cooperation length $L_c = c\tau_c$, the condition $k\tau_c \gg 1$, since $k = c/L$, is equivalent also to $L \ll L_c$.

A very remarkable fact is that Eq. (3.62) is identical to Eq. (3.45) for self-induced transparency if in the latter we replace θ by $\tilde{\theta}$ and β' by $1/\tau_R$. Hence to solve Eq. (3.62) we follow the same steps as for Eq. (3.45). In the case of Eq. (3.62) the solution expressed in terms of the variable x reads

$$x(t) = A\mathrm{e}^{t/\tau_R}, \tag{3.64}$$

and, by imposing the initial condition $\theta(0) = \theta_0$, which implies that $x(0) = \tan(\theta(0)/4) = \tan\theta_0/2$, we obtain

$$x(t) = \mathrm{e}^{(t-t_D)/\tau_R}, \qquad t_D = -\tau_R \ln\left(\tan\frac{\theta_0}{2}\right), \tag{3.65}$$

where we have introduced the time t_D, which corresponds to the delay of the emission peak with respect to the initial time $t = 0$. Next, exactly as in Eq. (3.50), we arrive at the result

$$\sin\left(\frac{\tilde{\theta}}{2}\right) = \sin\theta = \operatorname{sech}\left(\frac{t - t_D}{\tau_R}\right), \tag{3.66}$$

from which we conclude that the quantity $\Omega^2(t)$, which is proportional to the emitted intensity, is given by

$$\Omega^2 = \dot{\theta}^2 = \left(\frac{\dot{\tilde{\theta}}}{2}\right)^2 = \frac{1}{{\tau_R}^2}\left[\operatorname{sech}\left(\frac{t - t_D}{\tau_R}\right)\right]^2. \tag{3.67}$$

In the case of superradiance $\theta_0 = \pi/2$ and $t_D = 0$, so the emission is maximum at the start of the process, whereas in the case of superfluorescence there is initially a lethargic stage and the emission peak is located with a delay $t_D = -\tau_R \ln(\tan((1/2)\sqrt{2/\mathcal{N}})) \approx -\tau_R \ln(1/\sqrt{2\mathcal{N}}) = (\tau_R/2)\ln(2\mathcal{N})$ (see Fig. 3.4(a)). The most remarkable feature that emerges from Eq. (3.67) is that the peak height is proportional to τ_R^{-2}, which is proportional to the square of the atomic density N, as follows from Eq. (3.63), owing to the fact that $\tau_c = \beta^{-1}$ and Eq. (3.38). Correspondingly, the width of the emission peak is on the order of τ_R and therefore is inversely proportional to the atomic density. We must also note that, even if Eq. (3.62) for pure superradiance/superfluorescence is identical in form to Eq. (3.45) for self-induced transparency, there are basic differences between the two cases. The first main difference is that in the case of self-induced transparency, see Eq. (3.51), Ω^2 is proportional to β'^2 and therefore to β^2, hence on the basis of Eq. (3.38) the peak is simply proportional to the atomic density N and not to the square of the atomic density as in superradiance/superfluorescence. The second difference is that in the case of superradiance/superfluorescence, as a consequence of the fact that $r_3 = \cos\theta$ and of Eq. (3.66), the

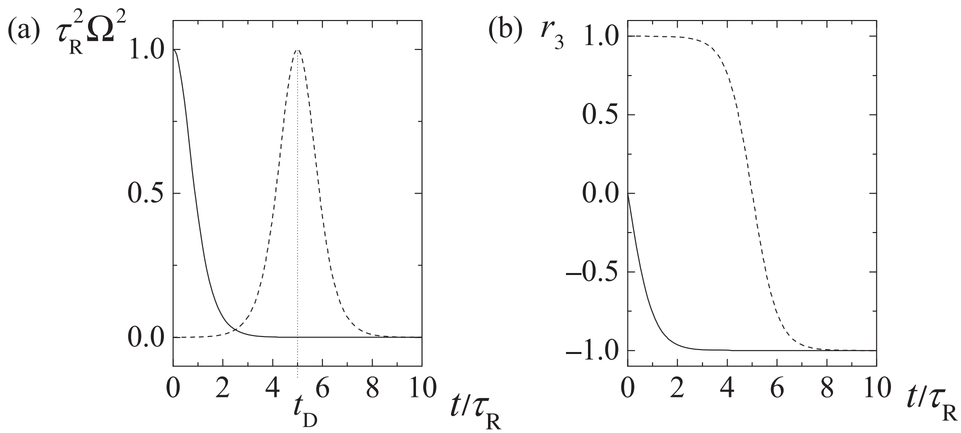

Figure 3.4 Pure superradiance/superfluorescence ($k\tau_c \gg 1$). (a) The time evolution of the emitted intensity in the case of superradiance ($\theta_0 = \pi/2$) and in the case of superfluorescence ($\theta_0 = 2e^{-5}$). (b) The corresponding time evolution of the population inversion. The solid curves correspond to superradiance, the broken curves to superfluorescence.

time evolution of the population inversion is given by (see Fig. 3.4(b))

$$r_3 = -\tanh\left(\frac{t - t_D}{\tau_R}\right), \tag{3.68}$$

which is quite different from the expression (3.54) for self-induced transparency. The difference arises from the fact that Eq. (3.62) is for $\tilde{\theta}$ whereas Eq. (3.45) is for θ. Hence in the superradiant/superfluorescent emission the atoms decay cooperatively in a time τ_R that is inversely proportional to the number of atoms. Pure superfluorescence was experimentally observed for the first time by Vrehen and Gibbs [39].

Let us now reduce the product $k\tau_c$ to values of order unity, which means also $k\tau_R$ of order unity, and is equivalent to the condition $L \approx L_c$. In this situation the pendulum is no longer overdamped and approaches the stable equilibrium state $\theta = \pi$ with some oscillations (*oscillatory superradiance/superfluorescence* [40]). Correspondingly, the emission intensity exhibits a number of peaks. After each peak, except the last one, the light is partially reabsorbed by the atoms, so the emission becomes partially reversible.

The time scale is still τ_R and this implies that the emitted intensity, which is proportional to $\Omega^2 = \dot{\theta}^2$, scales as the square of the atomic density. The height of the highest peak is lower than that of the pure superradiant/superfluorescent peak, and decreases when $k\tau_c$ is decreased. If we introduce the normalized time $t_1 = t/\tau_R$, Eq. (3.59) reads

$$\frac{d^2\theta}{dt_1^2} = (k\tau_c)^2\left(-\frac{d\theta}{dt_1} + \sin\theta\right). \tag{3.69}$$

Figures 3.5(a) and (b) show the time evolution of Ω^2 and of r_3, respectively, for $k\tau_c = \sqrt{0.1}$. The first experimental observation of oscillatory superfluorescence was reported by Feld and his collaborators [41].

Finally, let us further reduce the product $k\tau_c$ to values much smaller than unity, which is equivalent to assuming that $L \gg L_c$. In this configuration the radiation is emitted in a

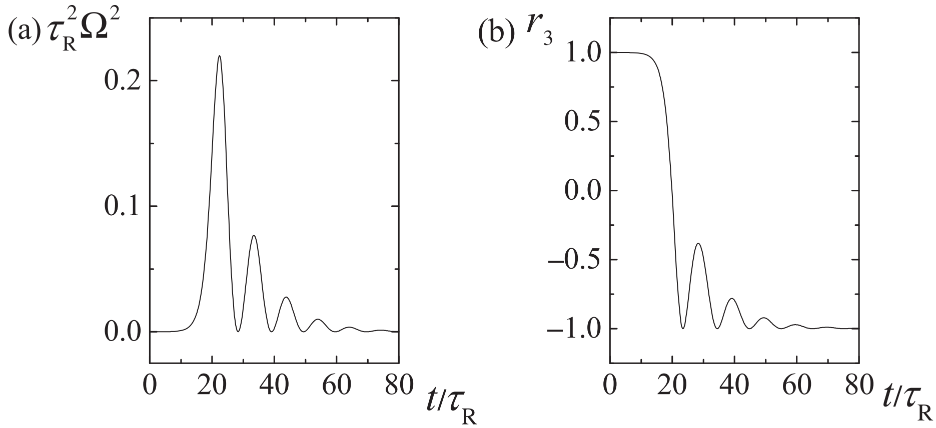

Figure 3.5 Oscillatory superfluorescence for $k\tau_c = \sqrt{0.1}$. (a) The emitted intensity. (b) Population inversion. The value of θ_0 is $\theta_0 = 2e^{-5}$.

very slowly damped sequence of peaks. The time scale is no longer τ_R but τ_c and, as a consequence, Ω^2 scales as the atomic density and no longer as the square of the atomic density, hence the emission is no longer superradiant/superfluorescent. If we introduce the normalized time $t_2 = t/\tau_c$, Eq. (3.59) takes the form

$$\frac{d^2\theta}{dt_2^2} + (k\tau_c)\frac{d\theta}{dt_2} = \sin\theta. \tag{3.70}$$

In the limit $k\tau_c = 0$, i.e. when L/L_c approaches infinity [42], this equation reduces to a pure pendulum equation. Its solutions correspond to an elliptic function with a period on the order of τ_c. The radiation is periodically emitted and reabsorbed in a reversible way. The quantum model, of which Eq. (3.70) in the limit $\tau_c = 0$ represents the semiclassical approximation, was formulated by Tavis and Cummings [43]. It describes the reversible dynamics of $\mathcal{N}$ atoms interacting with a cavity mode.

In the case of superfluorescence quantum noise is important not only in the initiation stage of the emission, but also throughtout the time evolution, and gives rise to anomalously large fluctuations, so that only a fully quantum picture is adequate.

The paradigm for the atom–field interaction at the quantum level is given by the Jaynes–Cummings model [44], which governs the reversible exchange of energy between a two-level atom and a cavity mode in the absence of any irreversible process. The development of the field of *cavity quantum electrodynamics* during the 1980s, with the realization of microcavities with strong coupling between a single atom and a cavity mode and an extremely small rate of escape of photons from the cavity, led to the realization of physical conditions corresponding to the Jaynes–Cummings model [45, 46].